

Instructor: ..... Date: .....

Learner: ..... Regional Branch: .....

Program: ..... Center: .....

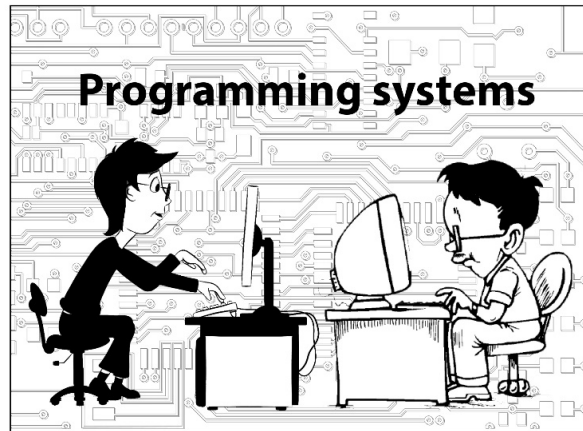
### CONTENT

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### Let's Start!

#### RUP and UML as Software Tools



#### Learning Outcome:

Describe common activities at work to a visitor using short sentences in familiar spoken and written situations.  
Talk about particular topics related to software programming.

#### Can Do Statements:

- I can compare RUP and UML.
- I can talk about the differences of both, RUP and UML.
- I can define both, RUP and UML at a general concepts.

#### Ask your partner:

- What are the tools to design object-oriented systems?
- Do you think organizations should get software that can adapt to their needs? Why?





### Let's Explore!

Laura's team works with new software development tools!



*Instructions: Read the text below.*

- Sara:** Hi Laura, how is your company working now?
- Laura:** well, we decided to acquire a framework for software development with adaptable methodologies, we also bought a language for object modeling. The first one is RUP and the standardized language is UML.
- Sara:** How would you define each one of them? Isn't your staff used to programming using those tools?
- Laura:** As a matter of fact, they are learning. They're just realizing that the unified modeling language or UML is a standardized graphical modeling language, which can be used to illustrate different issues about any project or system. And the rational unified process or RUP is a process framework for an organization to choose the way of elaborating their own software.
- Sara:** And, which one of those tools do you recommend the most?
- Laura:** Well, I'm not used to coding either in RUP or UML, however, making a comparison between them, I would say that UML offers a general-purpose modeling language to display an abstract model of a system thanks to visual artifacts. While RUP is not a single concrete prescriptive process but an adaptable framework used by organizations.
- Sara:** It seems your team is not used to working with those software tools yet. I think both of them have the same importance. Then, what is the main difference between RUP and UML?
- Laura:** In fact, the staff is not used to changing framework continuously, however they agreed to use it, once they saw the advantages, such as the tools they can bring for the company's production. About the difference, I can tell you that RUP provides you with a variety of practices to pick out and build specific types of software to start a development process, and UML appears to facilitate many of such practices, mainly during the elaboration phase.



### Let's Practice!

What is RUP and UML?

*Instructions: Circle the correct option to complete the dialogue.*

**Sara:** Hi, how is your (team / company / staff) working now?

**Laura:** well, we decided to acquire a (online / webpage / framework) for software development with adaptable methodologies, we also bought a language for object modeling. The first one is RUP and the standardized language is UML.

**Sara:** How would you define each one of them? Isn't your staff used to (programming / program / programmed) using those tools?

**Laura:** In fact, they are learning. They just realized that UML is a standardized graphical modeling language which you can use to illustrate different issues about any project or system. And RUP is a process framework for an (crew / panel / organization) to choose the way of elaborating their own (software / blackboard / outlaw).



### Let's Practice!

And which one of those tools do you recommend the most?

*Instructions: Use the words from the word bank to complete each sentence.*

explaining / to comparing / coding

1. I'm not used to \_\_\_\_\_ in any of both tools, RUP or UML.
2. I'm not used \_\_\_\_\_ but I would say UML offers a general-purpose modeling language while RUP is not a single concrete prescriptive process.
3. She's not used to \_\_\_\_\_ but she could tell you that UML allows to display an abstract model of a system thanks to visual artifacts, while RUP is an adaptable framework used by organizations.



## Extension Activity

UML and RUP for beginners

*Instructions: Complete the following steps. Fill in the Self-Assessment.*

- Look for more information about UML and RUP, and their tools.
- Identify the main functionalities of each tool.
- Create a brochure to explain how these tools work.
- Give examples in your explanation.
- Share your brochure with a partner.

### Self-Assessment

- |                                                         | Yes                   | No                    |
|---------------------------------------------------------|-----------------------|-----------------------|
| 1. I look for more information about UML and RUP.       | <input type="radio"/> | <input type="radio"/> |
| 2. I identify the main functionalities of each tool.    | <input type="radio"/> | <input type="radio"/> |
| 3. I create a brochure to explain how these tools work. | <input type="radio"/> | <input type="radio"/> |
| 4. I give examples.                                     | <input type="radio"/> | <input type="radio"/> |
| 5. I share my brochure with a partner.                  | <input type="radio"/> | <input type="radio"/> |



## Target Words

- **graphical modeling** (noun): the way of encoding a graph based representation over a multi-dimensional space, expressing a conditional dependence to visualize the domain model  
e.g. UML is a standardized graphical modeling language, which can be used to illustrate different issues about any project or system.
- **abstract model** (noun): a model that represents an extension of the reality  
e.g. UML offers a general-purpose modeling language to display an abstract model of a system.
- **prescriptive** (adjective): expressed as exactly as it is, that expresses literally how a process should be developed  
e.g. RUP is not a single concrete prescriptive process but an adaptable framework.
- **software development** (noun): a process to program a computer in order to get a software product  
e.g. We decided to acquire a framework for software development with adaptable methodologies.
- **pick out** (phrasal verb): to gather, to select from a group  
e.g. About the difference, I can tell you that RUP provides you with a variety of practices to pick out and build specific types of software to start a development process.



## Learning Points

### Be Used To + Ing In Negative

You use **used to** when you are accustomed to something or you don't find it unusual.

The expression **be used to** is sometimes followed by the gerund – the **ing** form of a verb.

With **be used to** in negative you usually talk about something that is not familiar to us or easy for us.

For negative sentences use this structure:

Subject + verb **be** (am, is, are) + **not** + **used to** + **verb** + **ing** + complement.

Example:

*I'm **not used to programming** in both, RUP and UML.*

*He **isn't used to modeling** with UML tools.*

*I **am not used to working** with adaptable software.*

Be used to can be used with contractions:

*He **is not used** to designing systems through UML.*

*He **isn't used** to designing systems through UML.*

Note: Remember that as they are expressions, be used to and get used to, will always be followed by a verb with ing.

### Examples

- *Isn't your staff used to programming using those tools?*
- *Well, I'm not used to coding either in RUP or UML.*
- *Your team is not used to working with those software tools yet.*
- *The staff is not used to changing framework continuously, however they agreed to use it, once they saw the advantages.*



## Learning Points

### Collective nouns vs Compound nouns

**Collective nouns** are the ones which name a group of individuals or a number of animals, places, things, objects, concepts and others, while the **compound nouns** are the result of combining two words.

| Collective nouns                                                                                                                                                                                                                                                                                      | compound nouns                                                                                                                                                                                                                                                                                                                                      |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Name a group                                                                                                                                                                                                                                                                                          | Merge two or three words                                                                                                                                                                                                                                                                                                                            |
| Some common collective nouns that are considered as a singular when they represent an entity: <ul style="list-style-type: none"> <li>• audience</li> <li>• team</li> <li>• jury</li> <li>• company</li> <li>• committee</li> <li>• crowd</li> <li>• class</li> <li>• group</li> <li>• team</li> </ul> | Observe some common compound nouns and the kind of part that compound those: <ul style="list-style-type: none"> <li>• software (adjective – noun)</li> <li>• input (adverb - verb)</li> <li>• spreadsheet (verb-noun)</li> <li>• blackboard (adjective – noun)</li> <li>• framework (noun- verb)</li> <li>• coding machine (gerund-noun)</li> </ul> |
| If you think of such noun as a group that is acting individually you can use a plural verb: <ul style="list-style-type: none"> <li>• <i>The staff are working in different departments.</i></li> </ul>                                                                                                | You can write compound nouns in different ways: <ul style="list-style-type: none"> <li>• separate words (post office)</li> <li>• words linked by a hyphen (check-in)</li> <li>• one word (notebook)</li> </ul>                                                                                                                                      |

### Examples

- *How is your company working now?*
- *The staff is not used to changing framework continuously.*
- *Choose the way of elaborating their own software.*
- *It seems your team is not used to working with those software tools yet.*
- *UML offers a general-purpose modeling.*



## Learning Points

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### RUP vs UML

UML is the unified modeling language to structure and visualize the design of a software system while RUP is an adaptable process framework to develop software.

The **UML** has its roots in the object-oriented methods, it appeared as a standardized specification language for object modeling. Since then, it has been considered a general-purpose modeling language which contains many artifacts and language constructs as graphical notation.

The **URP** is a development process framework that does not imply a prescriptive but an adaptable process which means an advantage for companies that have the possibility to select the appropriate elements of the process to their needs.